The Normal Distribution

ANSWER KEY

Section 1: The Empirical Rule (68-95-99.7)

1 In a normal distribution with mean $\mu = 100$ and $\sigma = 15$, approximately what percentage of scores fall between 70 and 130?

a. 95% ✓

b. 68%

c. 99.7%

d. 50%

$70 = 100 - 2(15)$ and $130 = 100 + 2(15)$. By the Empirical Rule, approximately 95% of data in a normal distribution falls within 2 standard deviations of the mean.

2 Using the 68-95-99.7 rule, what percentage of observations in a normal distribution fall above $\mu + 1\sigma$?

a. 16% ✓

b. 32%

c. 34%

d. 2.5%

Within 1 standard deviation is 68%. The remaining 32% is split evenly between the two tails: $\frac{32\%}{2} = 16\%$ in the upper tail.

3 If heights of adult men are $N(70, 3)$ inches, what percentage are taller than 76 inches?

a. 2.5% ✓

b. 5%

c. 16%

d. 0.15%

76 is $70 + 2(3) = \mu + 2\sigma$. Beyond 2 standard deviations is $100\% - 95\% = 5\%$. In the upper tail alone: $\frac{5\%}{2} = 2.5\%$.

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Section 2: Finding Normal Probabilities

4 For a standard normal variable Z, find $P(Z < -1.28)$ given that the standard normal CDF at -1.28 is approximately 0.10.

a. 0.10 ✓

b. 0.90

c. 0.20

d. 0.05

By definition of the cumulative distribution function, $P(Z < -1.28) = 0.10$.

5 If $P(Z < 1.96) \approx 0.975$, what is $P(-1.96 < Z < 1.96)$?

a. 0.95 ✓

b. 0.975

c. 0.05

d. 0.025

By symmetry, $P(Z < -1.96) = 1 - 0.975 = 0.025$. Then $P(-1.96 < Z < 1.96) = 0.975 - 0.025 = 0.95$.

6 A machine fills cereal boxes normally with $\mu = 16.2$ oz and $\sigma = 0.1$ oz. What proportion of boxes contain less than the stated label of 16.0 oz?

a. $P(Z < -2.0) \approx 0.0228$ ✓

b. $P(Z < -1.0) \approx 0.1587$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.5000

$z = \frac{16.0 - 16.2}{0.1} = \frac{-0.2}{0.1} = -2.0$. The tail probability is $P(Z < -2.0) \approx 0.0228$ (about 2.28%).

Section 3: Inverse Normal Calculations (Finding Values from Probabilities)

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7 Test scores are normally distributed with $\mu = 500$ and $\sigma = 100$. What score corresponds to the 90th percentile ($z \approx 1.28$)?

a. 628 ✓

b. 590

c. 600

d. 696

$x = \mu + z\sigma = 500 + 1.28(100) = 500 + 128 = 628$.

8 In a normal distribution, what are the z-scores corresponding to the first quartile (Q_1) and third quartile (Q_3)?

a. $z \approx -0.67$ **and** $z \approx +0.67$ ✓

b. $z \approx -1.00$ and $z \approx +1.00$

c. $z \approx -0.50$ and $z \approx +0.50$

d. $z \approx -1.96$ and $z \approx +1.96$

Q_1 has 25% of area below it, giving $z \approx -0.674$. Q_3 has 75% below, giving $z \approx +0.674$.

9 The weights of bags of potato chips are $N(\mu, 0.4 \text{ oz})$. If the heaviest 5% of bags weigh more than 16.66 oz ($z_{0.95} = 1.645$), find the mean weight μ.

a. 16.0 oz ✓

b. 17.3 oz

c. 15.8 oz

d. 16.2 oz

$x = \mu + z\sigma \Rightarrow 16.66 = \mu + 1.645(0.4) = \mu + 0.658 \Rightarrow \mu = 16.66 - 0.658 \approx 16.0$ oz.

Section 4: Assessing Normality with Normal Probability Plots

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10 What appearance in a Normal Probability (Q-Q) Plot indicates that the sample data comes from a roughly normal population?

a. **The plotted points follow a roughly linear pattern ✓**

b. The plotted points form a bell-shaped curve

c. The plotted points are completely horizontal

d. The plotted points form a curved parabolic shape

If sample data comes from a normal population, the normal quantile plot displays a straight linear trend.

11 If a Normal Probability Plot shows a clear upward curve (concave up), what does this indicate about the distribution?

a. **The distribution is skewed to the right ✓**

b. The distribution is skewed to the left

c. The distribution is perfectly symmetric

d. The distribution has no spread

A systematic concave-up bend in a normal probability plot indicates right-skewness in the sample data.

12 What is the total area under any valid probability density curve?

a. `1` ✓

b. 100

c. 0.5

d. Depends on the standard deviation

By the axioms of probability, the total area under any continuous probability density curve over its entire domain is exactly `1`.

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